Access to Insurance: Evidence from China's Migrant Population

Elsa Zhou

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Prof: Marcella Alsan
TA: Nicholas Scott-Hearn

1 Introduction

The impact of access to health insurance continues to be a strongly debated question, with direct policy-relevance for countries all across the globe building or reforming social service programs to meet the needs of their citizens. Insurance traditionally seeks to smooth consumption and income across time periods to accommodate uncertainty about future states, including changes in health, but is increasingly evaluated with a dual mandate of improving health outcomes in the future. Our class readings and discussions have touched on only one small part of the large body of evidence arrayed to answer the question of what health insurance really provides [1] [2] [3] [4].

China's unique social service distribution system presents a policy environment in which to study this question. The *hukou* identification system, assigned at birth, limits the distribution of social services such as health insurance, poverty assistance, unemployment and pension benefits to the hometown and urban/rural designation of the potential recipient [5]. In practice, this means that the hundreds of millions of rural-born workers that migrate to cities, called China's 'floating population,' do not have access to a wide array of social services [6]. The *hukou* system thus functions as an institutional mechanism that translates geographic origin into durable social inequality. This vestigial feature of China's socialist roots is recognized as a serious issue, but attempts at reform persistently straggle as local governments face enormous levels of debt and repayments on interest alone are already overwhelming.

China's health insurance system prior to 2016 was highly fragmented along the urban-rural divide that the *hukou* system enforces. Three separate programs operated in parallel: the Urban Employee Basic Medical Insurance (UEBMI), a contributory scheme for formally employed urban workers financed through payroll deductions; the Urban Resident Basic Medical Insurance (URBMI), a subsidized voluntary scheme for urban residents without formal employment; and the New Rural Cooperative Medical Scheme (NRCMS), a heavily subsidized voluntary program introduced in the early 2000s for rural residents [7]. Migrants (i.e. people with rural *hukou* living in urban areas) were essentially left out of receiving urban health insurance, ineligible for URBMI due to their rural *hukou* status and predominantly ineligible for UEBMI because of the prevalence of employment in informal labor [8]. Thus, prior to 2016, migrants either remained enrolled in the NRCMS rural scheme which did not distribute services in the cities where they lived, or go uninsured.

Recognizing the need for reform, the central government expanded access to health insurance in 2016 by merging URBMI and NRCMS into a unified Urban and Rural Residents Basic Medical Insurance (URRBMI) [10]. This created a pooled scheme, including accessibility to all urban residents regardless of *hukou* and thus extending health insurance to migrants for the first time. This insurance expansion to migrants after 2016 serves as a natural experiment for the impacts of access of health insurance. This paper uses a two-way fixed-effects differences-in-differences (DiD) strategy to estimate the causal impact of health insurance on health outcomes, healthcare utilization, labor market behavior, and migration patterns.

I find that the merger produced a counterintuitive immediate effect: rather than expanding coverage, migrants experienced a net increase in uninsurance of approximately 5.6 percentage points in the short run. This pattern is consistent with

administrative transition frictions, i.e. individuals disenrolling from NRCMS upon announcement of the new scheme but facing delays or procedural barriers in completing enrollment in URRBMI. Downstream effects on health utilization (doctor visits, hospitalization, self-reported illness) and self-rated health are statistically indistinguishable from zero, consistent with the lack of urban insurance take-up. Heterogeneity analysis reveals that low-income migrants experienced both smaller uninsurance increases and significant increases in hospitalization relative to middle-income counterparts, potentially pointing to important variances in demand for healthcare by income. Finally, the merger does not appear to have induced significant occupational resorting into high-risk employment or accelerated rural-to-urban migration within the two-year post-reform window.

Identifying the impact of expanded health insurance access is by itself an important question, but the specific population gaining access here is also key. Migrants constitute one of China's most structurally disadvantaged populations, facing compounding exclusions from the social services. For example, while migrant children are formally entitled to public schooling in their city of residence, in practice they face enrollment quotas, documentation requirements, and fees that effectively push most back to their home provinces entirely, separated from their parents [11]. These tens of millions of children are known as the 'left-behind children,' and this educational exclusion adds onto the health exclusion studied here. Families that are denied both adequate schooling and healthcare face present hardship and future penalties as they have diminished capacity to invest in the next generation's human capital.

As referenced earlier as well, migrants are disproportionately concentrated in the informal labor sector, in work like food delivery and ride-hailing, and earn substantially less than their urban counterparts while remaining ineligible for formal contracting that could give them access to employer-provided health insurance [8]. This income gap means that out-of-pocket health expenditures represent a proportionally larger financial shock for migrants, making the insurance coverage question not merely one of access but of serious financial protection against potentially unbearable costs. The heterogeneity results in this paper focus on this: it is precisely low-income migrants who show the largest hospitalization response following the merger, consistent with latent demand from migrants with the largest potential financial burdens.

Finally, the equity stakes here are relevant to China's macroeconomic trajectory. As China seeks to escape the middle-income trap and transition from low-cost manufacturing toward higher-value production, upgrading the human capital of its enormous migrant workforce is not optional. Persistent exclusion from health insurance and other social services depresses the productive potential of hundreds of millions of workers and their children, concentrating the gains of urbanization among a smaller urban elite [9]. In this sense, the 2016 merger is not only a health policy intervention but a test of whether China's institutional architecture can undergo transformative reform once again to serve its entire population. The evidence in this paper suggests that the administrative and political barriers to reform are formidable.

The remainder of the paper proceeds as follows. Section 2 describes the data used. Section 3 outlines the empirical strategy. Section 4 presents the main results and heterogeneity analyses. Section 5 provides robustness checks. Section 6 concludes.

2 Data

I use data from the China Family Panel Studies (CFPS) from 2010-2018, a large-scale longitudinal survey project launched by the Institute of Social Science Survey at Peking University starting in 2010, with surveys conducted biennially [12]. The extensive dataset covers economic activities, education, family dynamics, migration, and physical and mental health of individuals, families, and communities using survey responses from 16,000 households across 25 provinces.

The following table shows summary statistics for the full sample, and then subgroups representing migrants and urban natives.

Table 1 Summary Statistics

	(1) Full Sample		(2) Migrants		(3) Urban natives	
	mean	sd	mean	sd	mean	sd
Age	45.127	17.620	43.878	16.758	47.506	16.921
Gender (1 = Male)	0.496	0.500				
Ethnicity (1 = Han)	0.917	0.276				
Marital status (1 = Never, 2 = Married, 3 = Cohabit, 4 = Divorced, 5 = Widowed)	2.070	0.866	2.048	0.835	2.101	0.872
logincome	9.492	1.435	9.593	1.341	9.964	1.155
Share of UEBMI	0.118	0.322	0.066	0.248	0.386	0.487
Share of URBMI	0.075	0.263	0.033	0.180	0.259	0.438
Share of NRMCS	0.621	0.485	0.773	0.419	0.097	0.296
Share uninsured	0.102	0.302	0.107	0.309	0.156	0.363
Felt physical discomfort in the past 2 weeks	0.294	0.456	0.279	0.448	0.283	0.450
Have you seen a doctor	2.136	1.804	2.205	1.835	2.416	1.913
Hospitalized last year due to illness/injury	0.102	0.302	0.102	0.303	0.103	0.304
Self-rated health status	2.798	1.291	2.752	1.261	2.789	1.206
Observations	180712		39149		38983	

Table 1 presents summary statistics for the full CFPS sample and separately for migrants and urban natives. The full sample of 180,712 observations has a mean age of 45, is roughly gender-balanced (49.6% male), and is predominantly Han Chinese (91.7%). The insurance distribution reveals the stark pre-merger coverage landscape that motivates this study: NRCMS (the rural scheme) accounts for the largest share of coverage in the full sample (62.1%), while UEBMI and URBMI combined (the urban schemes) cover only 19.3%. Breaking this down by group makes the inequality in coverage more pronounced. Migrants are overwhelmingly enrolled in NRCMS (77.3%) and have very low rates of urban insurance: only 6.6% hold UEBMI and 3.3% hold URBMI, compared to 38.6% and 25.9% respectively among urban natives. Notably, urban natives have a higher rate of uninsurance (15.6%) than migrants (10.7%), which likely reflects that some urban natives opt out of available schemes rather than facing structural exclusion from them. Health outcomes and utilization are broadly similar across the two groups — rates of physical discomfort, hospitalization, and self-rated

health are comparable between migrants and natives — which is consistent with a healthy migrant effect: migrants who select into urban areas may be positively selected on health relative to the general rural population.

Table 2 Pre-period Covariate Balance (Urban Sample, 2010-2014)

	(1)				
	Migrant	Native	Diff	SE	p-val
Age	46.941	44.047	2.894	0.155	0.000
Gender	0.494	0.471	0.023	0.005	0.000
Ethnicity	0.961	0.937	0.024	0.002	0.000
Marital status	2.093	2.065	0.028	0.008	0.000
logincome	9.814	9.280	0.534	0.017	0.000
Observations	47148				

Table 2 presents pre-period covariate balance between migrants and urban natives in the urban sample, restricted to the pre-merger waves (2010–2014). Every covariate differs significantly between the two groups at the 1% level. Migrants in the pre-period are slightly younger (46.9 vs. 44.0 years), more likely to be male (49.4% vs. 47.1%), more likely to be Han (96.1% vs. 93.7%), slightly more likely to be married, and have meaningfully higher log income (9.81 vs. 9.28). The income difference may point to some positive selection of migrants. Overall, the statistically significant differences across all covariates confirm that migrants and urban natives are not directly comparable on observable characteristics, which is why the two-way fixed effects design is necessary.

3 Methods

To estimate the causal effect of China's 2016 merger of urban and rural health insurance schemes on migrant health insurance coverage, healthcare utilization, and labor market outcomes, I employ a two-way fixed effects difference-in-differences design with the following form:

$$Y_{it} = \beta(\text{Migrant}_i \times \text{Post}_t) + \alpha_i + \gamma_t + \epsilon_{it} \tag{1}$$

where α_i and γ_t denote individual and year fixed effects respectively. Standard errors are clustered at the county level. The sample is restricted to urban residents (comprising urban natives and migrants), and Migrant_i is an indicator equal to one for individuals with rural *hukou* residing in urban areas, while Post_t is an indicator equal to one for survey years 2016 and after. The coefficient of interest β captures the difference-in-differences estimate: the differential change in outcomes for migrants relative to urban natives following the merger. In words, this regression asks whether

migrants, who gained health insurance coverage under the reform, experienced different changes in health and labor market outcomes than urban natives, after absorbing time-invariant individual characteristics and economy-wide annual trends.

The variation underlying this design comes from the timing of the 2016 merger interacted with migrants' pre-existing exposure to the urban-rural insurance gap. Because migrants holding rural *hukou* were previously enrolled in the New Rural Cooperative Medical Scheme (NRCMS), which did not provide coverage outside their home county, they faced a substantial barrier to accessing healthcare in their city of residence prior to the merger. Urban natives faced no such barrier. The merger plausibly removed this differential barrier in a manner that was common across provinces and occurred roughly at the same point in time, providing the identifying variation. The discrete timing of the policy is primarily motivated by the high level of political centralization in China's one-party system. The validity of the parallel trends assumption, or that migrants and urban natives would have followed similar outcome trajectories were it not for the policy, is assessed via event study estimates in Section 5.

In this paper, I estimate a second specification adding individual demographic covariates to the baseline:

$$\begin{aligned} Y_{it} = \beta(\text{Migrant}_i \times \text{Post}_t) \\ + \beta_1 \text{Age}_{it} + \beta_2 \text{Married}_{it} + \beta_3 \text{LogIncome}_{it} + \alpha_i + \gamma_t + \epsilon_{it} \end{aligned} \tag{2}$$

where standard errors are again clustered at the county level. Gender and ethnicity indicators are omitted, as these are time-invariant characteristics absorbed entirely by the individual fixed effects α_i. Thus, by using individual and year fixed effects, this specification controls for time-varying demographic characteristics that may confound the baseline estimate if they trend differentially between migrants and urban natives around 2016.

The analysis proceeds in two stages. In the first stage, I examine the effect of the merger on insurance uptake — specifically enrollment in UEBMI, URBMI, NRCMS, and the probability of being uninsured — to verify that the policy produced the intended change in coverage. In the second stage, I examine downstream effects on healthcare utilization outcomes including self-reported illness, doctor visits, hospitalization, and self-rated health. This two-stage structure follows the logic of a reduced-form causal chain: the merger first changed insurance coverage, which in turn changed healthcare-seeking behavior and health outcomes.

I further conduct heterogeneity analyses by augmenting the baseline specification with interactions between $\text{Migrant}_i \times \text{Post}_t$ and top- and bottom-income quartile indicators:

$$\begin{aligned} Y_{it} = \beta(\text{Migrant}_i \times \text{Post}_t) + \beta_L(\text{Migrant}_i \times \text{Post}_t \times \text{LowIncome}_i) \\ + \beta_H(\text{Migrant}_i \times \text{Post}_t \times \text{HighIncome}_i) \\ + \beta_1 \text{LowIncome}_i + \beta_2 \text{HighIncome}_i + \alpha_i + \gamma_t + \epsilon_{it} \end{aligned} \tag{3}$$

where LowIncome_i is an indicator for individuals below the 25th percentile of the log income distribution, and HighIncome_i is an indicator for individuals above the 75th percentile of the log income distribution. The purpose of this part of the analysis is to assess whether the effects of insurance access were concentrated among

economically vulnerable migrants, for whom the pre-2016 lack of access to health insurance represented a proportionally larger financial barrier.

Finally, I assess the parallel trends assumption underlying the difference-in-differences design by estimating event study specifications of the following form:

$$Y_{it} = \sum_{\tau \neq 2016} \beta_\tau (\mathbf{1}[\text{Year} = \tau] \times \text{Migrant}_i) + \alpha_i + \gamma_t + \epsilon_{it} \tag{4}$$

where τ indexes survey years $\{2010, 2012, 2014, 2018\}$ and 2016 serves as the omitted reference year. The one exception to this is in the labor market analysis, where I exclude 2010. The coefficients β_τ for $\tau < 2016$ test whether migrants and urban natives were on differential pre-trends prior to the merger, which is necessary to validate the DiD design. Failure to reject the joint hypothesis $H_0 : \beta_{2010} = \beta_{2012} = \beta_{2014} = 0$ is interpreted as supporting evidence for parallel trends. The coefficient β_{2018} captures the post-merger effect and should be consistent with the baseline DiD estimate $\hat{\beta}$.

4 Results

4.1 Insurance outcomes

Table 3 presents the first-stage estimates of the effect of the 2016 merger on insurance coverage. The coefficient of interest, *treat_post*, captures the differential change in each insurance outcome for migrants relative to urban natives after 2016. I find that migrants experienced a statistically significant decline in URBMI enrollment of 3.2 percentage points ($p < 0.05$) and a decline in NRCMS enrollment of 3.4 percentage points ($p < 0.05$), alongside a significant increase in the probability of being uninsured of 5.6 percentage points ($p < 0.01$). The probability of holding any urban insurance fell by 5.6 percentage points ($p < 0.01$).

Table 3 Effect of Merger on Insurance Uptake

	(1) UEBMI	(2) URBMI	(3) NRCMS	(4) Uninsured	(5) Any Urban Ins.
treat_post	-0.023	-0.032**	-0.034**	0.056***	-0.056***
	(0.017)	(0.016)	(0.014)	(0.015)	(0.018)
Obs	24631	24631	24631	24631	24631
Within R^2	0.004	0.002	0.002	0.005	0.004

Standard errors in parentheses
Two-way FE (individual & year). SE clustered at county.
Gender and ethnicity omitted: collinear with individual FE.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Taken together, these estimates suggest that while the merger was intended to expand urban coverage to migrants, the immediate measured effect was a net reduction in coverage. Migrants appeared to drop out of their rural insurance (NRCMS) without any corresponding uptake of urban insurance in its place. This pattern is consistent

with administrative transition frictions, whereby individuals disenroll from their prior scheme when a new program is announced but face delays or barriers in completing enrollment in the new one.

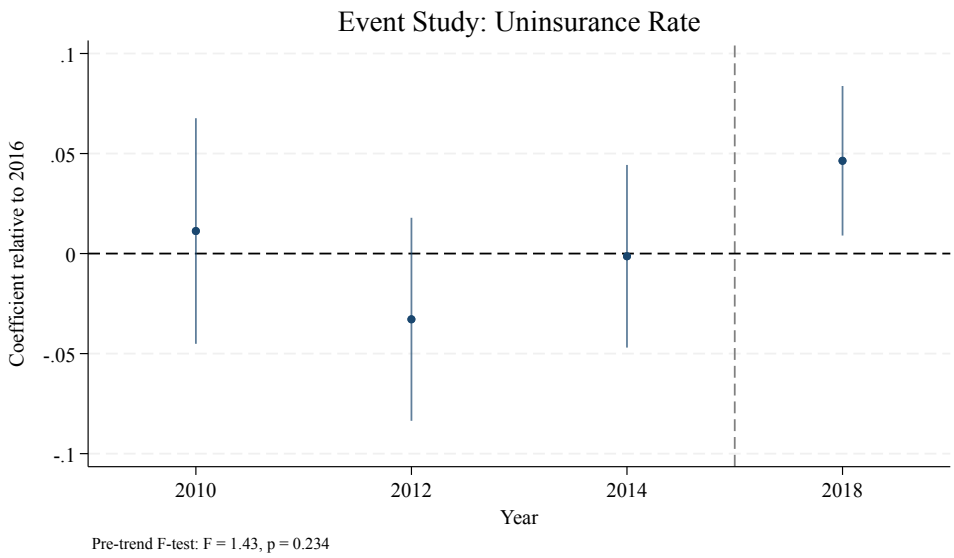

Fig. 1 Uninsurance

The event study for uninsurance (Figure 1) shows flat pre-period coefficients (F= 1.43, $p = 0.234$), supporting parallel trends for this outcome. The post-2016 coefficient on uninsurance is positive and consistent with the baseline DiD estimate, suggesting that the observed increase in uninsurance among migrants is attributable to the merger rather than pre-existing trends.

4.2 Health outcomes

Table 4 presents estimates of the merger's downstream effects on health. None of the four utilization or health outcomes - reported illness, doctor visits, hospitalization, or self-rated health - yields a statistically significant coefficient, and the point estimates are small in magnitude. The coefficient on seeing a doctor is 0.141 (SE = 0.244) and on self-rated health is -0.021 (SE = 0.041). Neither is distinguishable from zero. In addition, the within-R^2 values are near zero for all outcomes, indicating that the treatment interaction explains little of the within-individual variation in health after absorbing fixed effects and year trends.

Table 4 Effect of Merger on Health Outcomes

	(1) Sick	(2) See Doctor	(3) Hospitalized	(4) Self-Rated Health
treat_post	0.021	0.141	0.002	-0.021
	(0.020)	(0.244)	(0.014)	(0.041)
Obs	24188	3368	20468	24589
Within R2	0.000	0.002	0.000	0.001

Standard errors in parentheses
Two-way FE (individual & year). SE clustered at county.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

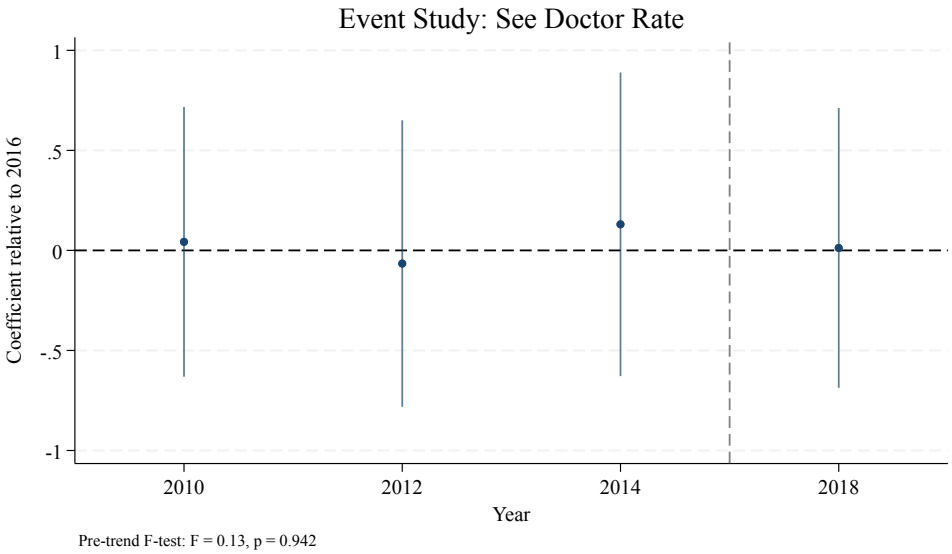

Fig. 2 Utilization: seeing a doctor

The event studies corroborate this null finding. Figure 2 shows flat and precisely estimated pre-period coefficients for doctor visits ($F = 0.13$, $p = 0.942$), providing strong support for parallel trends on this outcome, and a post-period coefficient consistent with zero. Figure 3 similarly shows no meaningful pre-trend or post-trend movement in self-rated health ($F = 1.47$, $p = 0.222$). The absence of detectable health effects is consistent with the first-stage finding of lacking up-take in urban insurance: if migrants experienced an increase in uninsurance rather than expanded coverage following the 2016 reform, I would not expect to observe improvements in healthcare utilization or health status in the short run.

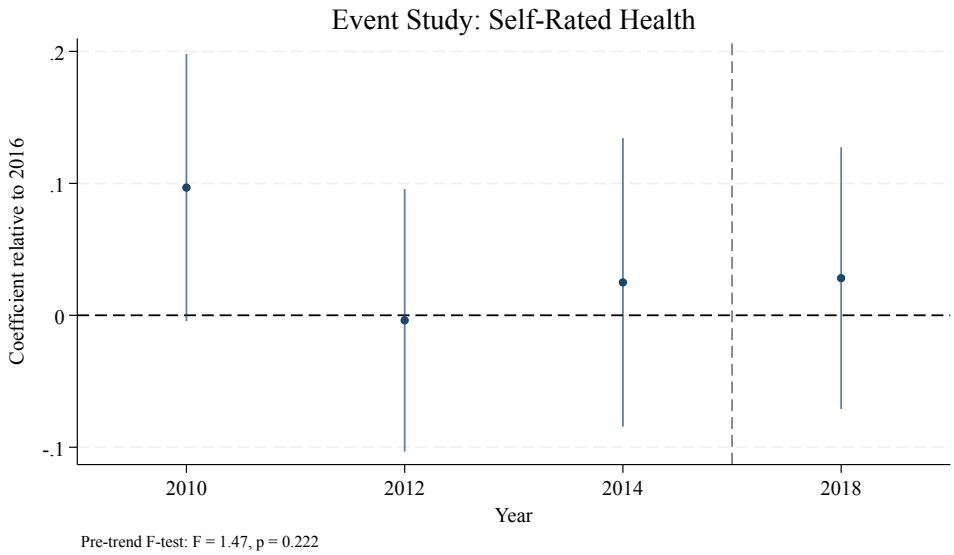

Fig. 3 Outcome: self-rated health

Figure 4 presents predicted facility-choice probabilities from a multinomial logit, separately for migrants and urban natives across the survey years. Because individual fixed effects are not feasible in multinomial logit, the figure controls for year-FE, age, gender, ethnicity, marital status, and log income. Among urban natives (labeled in the graph as migrant = 0), general hospital usage is consistently the dominant choice and appears stable over time. Among migrants, general hospital usage is lower on average but shows a modest upward trend, while usage of township and clinic facilities remain relatively flat. These patterns are descriptive rather than causal, but they weakly suggest that facility preferences of migrants may be shifting slightly toward higher-tier providers over time. This may also point to other frictions related to healthcare utilization beyond insurance.

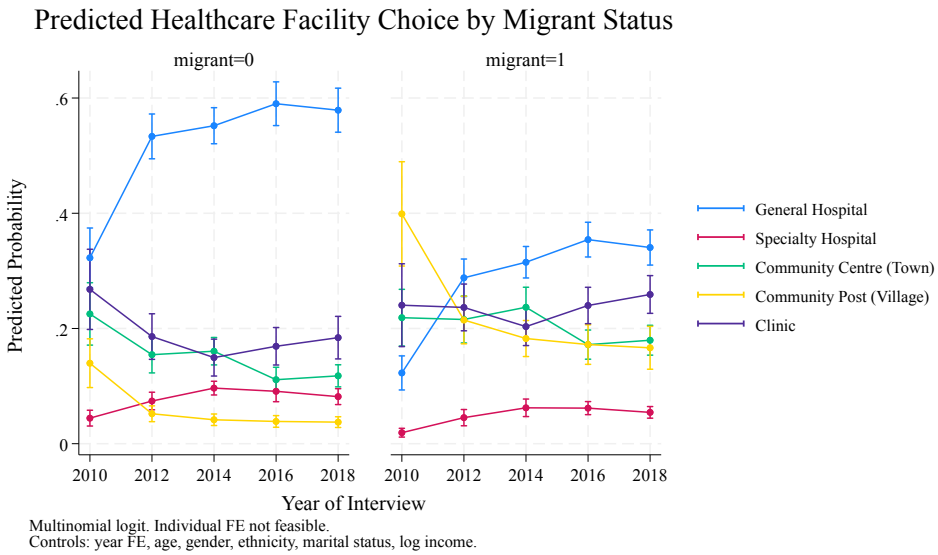

Fig. 4 Utilization: Medical facility of choice

4.3 Heterogeneity

Table 5 examines whether the merger's effects on insurance coverage, doctor visits, and hospitalization varied by income. The baseline *treat_post* coefficient on uninsurance is 0.075 ($p < 0.01$), indicating a 7.5 percentage point increase in uninsurance for middle-income migrants, a larger estimate than the pooled estimate in Table 3. Since the baseline *treat_post* reflects middle-income migrants, the negative interaction implies low-income migrants' uninsurance rose by $0.075 - 0.058 = 0.017$ percentage points, compared to 0.075 for the middle-income group. This is a somewhat counterintuitive finding. Low-income migrants might reasonably face greater barriers to enrollment in a new urban scheme, either from information asymmetries or administrative difficulties, yet the estimates suggest they were partially shielded from the uninsurance increase. One possible interpretation is that low-income migrants were more likely to be targeted by enrollment outreach programs or to qualify for subsidized premiums under the merged scheme.

For hospitalization, *treat_low* is 0.064 ($p < 0.05$), suggesting that low-income migrants experienced a significant increase in hospitalizations relative to middle-income migrants after the merger. This could reflect pent-up demand for inpatient care among the most economically vulnerable migrants who had previously been deterred by out-of-pocket costs. However, no significant heterogeneity is detected for doctor visits across income groups. It is worth noting that high-income migrants (*treat_high*) show no statistically significant differential response on any outcome, perhaps pointing to similarities between the middle- and high-income groups and the especially harsh conditions of the lowest-income quartile.

Table 5 Heterogeneous Effects by Income Quartile

	(1) Uninsured	(2) See Doctor	(3) Hospitalized
treat_post	0.075***	0.001	-0.007
	(0.018)	(0.303)	(0.016)
treat_low	-0.058**	0.316	0.064**
	(0.025)	(0.445)	(0.031)
treat_high	-0.025	0.270	0.004
	(0.016)	(0.420)	(0.020)
Obs	24631	3368	20468
Within R^2	0.007	0.004	0.001

Standard errors in parentheses
Two-way FE (individual & year). SE clustered at county.
treat_low (treat_high) = treat_post × bottom (top) income quartile.
Mid-income half is the omitted reference group.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

4.4 Labor market outcomes

Table 6 Effect of Merger on High-Risk Occupational Sorting (Baseline Barrier DiD)

	(1) In High-Risk	(2) Switch into H-Risk
treat_intensity	-0.011	0.050***
	(0.021)	(0.016)
Obs	15308	15308
Within R2	0.010	0.005

Two-way FE (individual & year). SE clustered at county.
`treat_intensity` = (uninsured/NRCMS at baseline) × post × migrant.
High-risk employment = construction, manufacturing, or mining (industry codes 2, 3, 5).
Sample: urban residents, 2012–2018. 2010 wave excluded due to sampling irregularities.
Parallel trends supported by event study (F = 0.00, p = 0.996).
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6 presents estimates of effect of the merged health insurance scheme on two occupational outcomes: the share of high-barrier migrants employed in high-risk sectors (column 1), and the share switching into high-risk employment from another sector over the course of the survey (column 2). High-risk employment is defined as construction, manufacturing, or mining, industries characterized by elevated physical hazard and workplace injury risk. The theoretical motivation traces similar reasoning to the standard VSL approach discussed in our class and engages with some part of moral hazard: if a migrant has no health insurance and gets injured at work, they bear the full out-of-pocket cost of medical care. This expected cost may deter migrants

from taking jobs in dangerous sectors in the first place, even if those jobs pay better. The 2016 merger, by providing insurance coverage to migrants, plausibly reduces this deterrent and enables greater occupational hazard and switching into these high-risk jobs.

To test this, the occupation analysis uses a slightly different design than the main DiD. Rather than comparing all migrants to all urban natives before and after 2016, the treatment variable here (*treat_intensity*) exploits variation within the migrant population in how exposed each individual was to the insurance gap prior to the merger. Specifically, migrants who were either uninsured or enrolled only in NRCMS at their first observed wave are classified as "high-barrier" migrants, meaning they faced the steepest insurance penalty before the reform. The smaller share of migrants who already had some form of urban insurance at baseline are classified as "low-barrier." The *treat_intensity* variable is then the interaction of this high-barrier indicator with the post-2016 indicator, estimated within the sample of those living in urban areas.

The event study for high-risk employment (Figure 5) shows flat pre-period coefficients with a pre-trend F-statistic of essentially zero ($F = 0.00$, $p = 0.996$), providing strong support for parallel trends on this outcome. The DiD estimate for high-risk employment is 0.011 (SE = 0.021), which is small and statistically indistinguishable from zero. Despite clean identification, the merger does not appear to have induced high-barrier migrants to sort into physically hazardous employment at detectable rates within the two-year post-merger window captured by the 2018 wave.

The switching outcome (column 2) shows different results. the DiD estimate is 0.050 ($p < 0.01$), suggesting that high-barrier migrants were significantly more likely to transition into high-risk employment after the merger relative to low-barrier migrants. However, the event study for this outcome (Figure 6) reveals a significant pre-trend violation ($F = 11.60$, p = 0.000), indicating that high- and low-barrier migrants were already on divergent switching trajectories before 2016. This undermines a causal interpretation of the column (2) estimate, and so the result should be treated as descriptive at best. The divergence between the two occupational outcomes likely reflects the noisier nature of this switching variable, which is sensitive to compositional changes in who is observed across consecutive waves.

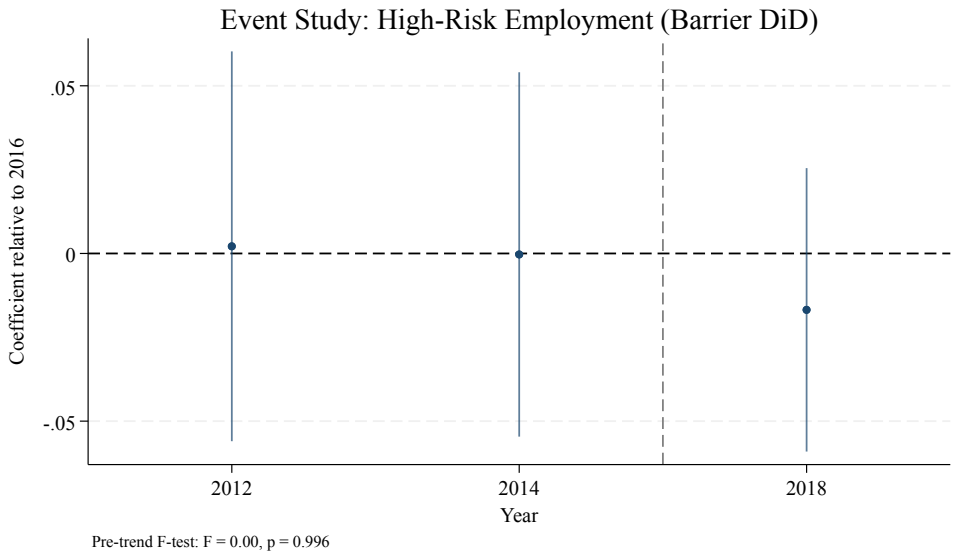

Fig. 5 Occupation: Rate of high-risk occupation

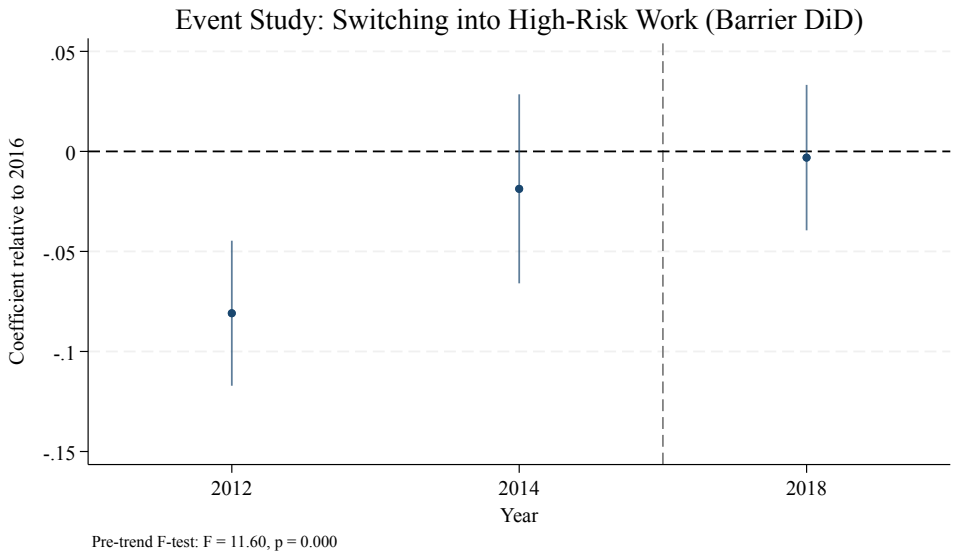

Fig. 6 Occupation: Switching into high-risk employment

4.5 Migration outcomes

Table 7 Effect of Insurance Merger on Migration (Baseline Barrier DiD)

	(1) Rural-to-Urban Migration
treat_intensity	-0.028
	(0.031)
Obs	15300
Within R2	0.003

Standard errors in parentheses
Standard errors in parentheses
Two-way FE (individual & year). SE clustered at county.
treat_intensity = (uninsured/NRCMS in 2010) × post-2016.
Sample: individuals observed rural in 2010.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7 examines whether the merger induced greater rural-to-urban migration, using the same barrier DiD design applied to individuals observed as rural residents in 2010. The estimated effect of *treat_intensity* on rural-to-urban migration is -0.028 (SE = 0.031), which is not statistically significant.

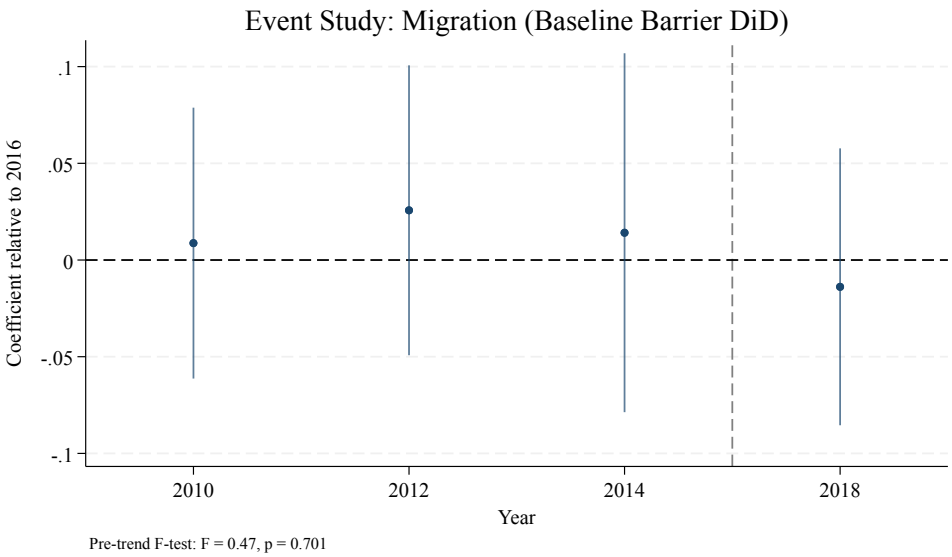

Fig. 7 Migration

The event study (Figure 7) shows flat pre-period coefficients ($F = 0.47$, $p = 0.701$), supporting the validity of the parallel trends assumption for this outcome. The null result on migration suggests that the 2016 merger did not meaningfully alter the incentive to migrate from rural to urban areas, at least within the two-year window captured by the 2018 wave.

5 Robustness checks

Section 5 presents four robustness checks on the main health outcome estimates from Table 4. Across all checks, the core finding is unchanged: the merger produced no statistically significant effect on illness, doctor visits, hospitalization, or self-rated health for migrants relative to urban natives. The null result is robust to a range of alternative specifications and sample restrictions.

Table 8 Robustness: Health Outcomes, Province-Level Clustering

	(1) Sick	(2) See Doctor	(3) Hospitalized	(4) Self-Rated Health
treat_post	0.021	0.141	0.002	-0.021
	(0.022)	(0.209)	(0.014)	(0.033)
Obs	24195	3368	20472	24598
Within R2	0.000	0.002	0.000	0.001

Standard errors in parentheses
Two-way FE (individual & year). SE clustered at province (N=31).
With few clusters, conventional SE may be downward-biased.
Main results (Table 4) cluster at county (N=550).
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

The main results cluster standard errors at the county level (N = 550). Table 8 re-estimates the same specification clustering at the province level (N = 31) instead. With only 31 clusters, conventional cluster-robust standard errors may be downward-biased, meaning the main results could in principle be understating uncertainty. In practice, the point estimates are identical to Table 4 by construction, and the standard errors are broadly similar - they are slightly larger for self-rated health (0.033 vs. 0.041) and slightly smaller for doctor visits (0.209 vs. 0.244). Similar to Table 4, no outcome crosses the threshold of statistical significance. The null result is therefore not an artifact of the county-level clustering assumption.

Table 4 controls for time-varying demographics (age, marital status, and log income) on the grounds that these may trend differentially between migrants and urban natives around 2016 and confound the baseline estimate. Table 9 drops these controls entirely, applying the two-way fixed effects design without any covariates. The sample expands substantially as a result (from roughly 24,000 to 67,000 observations) because observations dropped due to missing covariate values are restored. Point estimates for sick, doctor visits, and hospitalization remain small and insignificant. The one exception is self-rated health, which becomes significant at the 1% level

Table 9 Robustness: Health Outcomes, No Controls

	(1) Sick	(2) See Doctor	(3) Hospitalized	(4) Self-Rated Health
treat_post	0.013	0.075	0.004	-0.067***
	(0.012)	(0.086)	(0.009)	(0.020)
Obs	67891	14052	50597	72296
Within R2	0.000	0.000	0.000	0.000

Standard errors in parentheses
Two-way FE (individual & year). SE clustered at county.
No demographic controls included.
Main results (Table 4) include age, marital status, log income.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

(0.067, SE = 0.020) in the no-controls specification but is not significant in the main specification. This divergence suggests that the self-rated health estimate is sensitive to the inclusion of income and demographic controls, and that the apparent effect in the no-controls specification may reflect differential trends in income or marital status between migrants and natives rather than a true effect of the merger. The main controlled specification is preferred on these grounds, and the null results are still supported.

Table 10 Robustness: Health Outcomes, Excluding 2010 Wave

	(1) Sick	(2) See Doctor	(3) Hospitalized	(4) Self-Rated Health
treat_post	0.017	0.134	0.000	-0.003
	(0.023)	(0.281)	(0.014)	(0.048)
Obs	14957	1978	11271	15275
Within R2	0.001	0.004	0.002	0.001

Standard errors in parentheses
Two-way FE (individual & year). SE clustered at county.
Sample restricted to cyear ¿= 2012.
Main results (Table 4) include all waves 2010-2018.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

The 2010 wave is the inaugural CFPS survey year and is plausibly subject to greater sampling uncertainty than later waves. Table 10 restricts the sample to 2012–2018 to assess whether the main results are sensitive to the inclusion of this wave. All point estimates remain small and statistically insignificant, and are closely comparable in magnitude to those in Table 4. The null result on health outcomes does not depend on the 2010 wave.

As a further test of the validity of the DiD design, Table 11 assigns a fake treatment date of 2014 and re-estimates the main specification on the pre-merger sample only (2010–2014). If the DiD estimate in Table 4 reflects a genuine policy effect, there

Table 11 Robustness: Placebo Test, Fake Treatment Date 2014

	(1) Sick	(2) See Doctor	(3) Hospitalized	(4) Self-Rated Health
placebo_treat	0.015	0.226	0.021	-0.035
	(0.023)	(0.349)	(0.014)	(0.059)
Obs	14833	1871	14832	15168
Within R2	0.001	0.004	0.001	0.001

Standard errors in parentheses
Two-way FE (individual & year). SE clustered at county.
Sample restricted to 2010-2014 (pre-merger waves only).
placebo_treat = migrant × (cyear ≥ 2014).
Null result supports validity of main DiD estimate.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

should be no observable significant treatment effect with the fake pre-period treatment. Consistent with this expectation, the placebo estimates are small and statistically insignificant across all four outcomes. This null result provides additional supporting evidence that the main DiD design is not picking up pre-existing trends between migrants and urban natives.

6 Conclusion

This paper examines the causal effects of China's 2016 merging of the Urban Resident Basic Medical Insurance and New Rural Cooperative Medical Scheme into a unified scheme on the coverage, health, and labor market outcomes of migrants. Using a two-way fixed-effects difference-in-differences design with data from the China Family Panel Studies (2010–2018), I find that the reform produced a short-run increase in uninsurance among migrants of approximately 5.6 percentage points. This result points to administrative transition frictions, as migrants appear to have disenrolled from their rural scheme (NRCMS) without successfully completing enrollment in an urban scheme. The absence of detectable downstream effects on healthcare utilization or self-rated health follows logically from this first-stage finding. On the whole, these findings carry several important implications for health equity and policy design. They emphasize an important gap between the ambition of insurance expansion and the administrative capacity required to deliver it. The *hukou* system has long structured the exclusion of migrants from urban social services, but formally inclusive reforms can fail to reach intended beneficiaries if enrollment processes are complex or information asymmetries prevent an otherwise smooth transition. For an extremely vulnerable population like migrants, these administrative frictions are especially important to pay attention to in policy design and overcome. In addition, the heterogeneity results suggest that low-income migrants were partially shielded from the uninsurance increase and experienced significant increases in hospitalization. If targeted outreach or other intentional components of the 2016 reform drove this result, then policy-makers may

have succeeded in reaching the most extremely economically vulnerable first. Ideally, similar initiatives can be staggered to better reach the remainder of the migrant population.

This paper has several limitations. The two-year post-reform window captured by the 2018 wave may be insufficient to detect important components of longer-run effects of health insurance coverage expansion, such as chronic disease management, preventive care, or the full unfolding of labor market outcomes. The biennial survey structure also prevents analysis of the precise timing of enrollment transitions within the reform window. Future work with administrative enrollment data or higher-frequency survey data could better characterize the dynamics of coverage transition and the timeline over which health effects emerge. More broadly, the findings underscore that the design of enrollment mechanisms and the administrative infrastructure of insurance transitions are first-order determinants of reform effectiveness.

6.1 Further discussion

A paper exploring social service distribution and structural social inequality stemming from *hukou* must be accompanied by discussion of the governance structure in China. China's political structure can be characterized as highly centralized in key decision-making, but highly decentralized in implementation. That is, the central government faces a classic principal-agent problem, where they set the reform agenda but cannot fully observe whether local agents are successful in their rollout. Further, the dynamic between local governments and the central government is colored by the 1994 fiscal reform, where the central government decided that tax revenues would be split evenly between the two but that local governments would bear the majority of expenditures, including the responsibility to distribute social services [13]. The embedded local deficit was intended to be supplemented and fixed by transfers from the central government, but in practice, the imperfect system leaves local governments consistently underfunded. This prominently led to the expansion of land sales by local governments and the subsequent rise of the property market in China, unlocking an enormous economic engine and facilitating infrastructure investments at a staggering pace. However, since then, the property market has sagged, infrastructure projects and state-owned enterprises have generated smaller and smaller returns, and debt obligations have continued to rise for local governments [14]. This belt-tightening is accompanied by depressed ambition in local officials to innovate ruthlessly like in the earlier years of China's development, intimidated by Xi Jinping's harsh anti-corruption campaign [15]. Thus, in this context, investment in health insurance expansion easily falls by the wayside. Local officials are too focused on the more pressing issue of ballooning debt with no easy stream of revenue, and citizens in the authoritarian regime have no dedicated avenue to voice their complaints and petition for relief.

Further, corporations are not incentivized to deliver expanded employee insurance. A key component leading to China's current level of inequality, despite extremely high levels of state control over assets and decision-making [16], is that corporations currently stand to gain the most from gains in the economy at large (for example, profits from state-owned enterprises that are funded by taxpayers are directed back into their own corporate budgets, not the public budget) [17]. Offering expanded

insurance to employees produces little advantage for corporations in a labor market characterized by supply.

With this context in mind, and looking ahead to China's future, the proper implementation of expanded health insurance only becomes more important. China's population is ageing, and will only demand health care in increasing amounts [18]. The structural inefficiencies of China's system of social services distribution is extremely evident in health care. The ultimate need of the party to maintain control over all domains of life ensures that private provision of health care and insurance is formally a non-option. Still, public provision in China is by no means an easy arena. Competition characterizes almost every facet of the economy, to the point of overcapacity and inefficient duplication of industrial clusters even in industries with high fixed costs as local governments vie for market share. With the right subsidies or incentives, perhaps the government could target health care as the next sector to throw into the fierce arena of local government competition.

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